

MHF4U - Advanced Functions

Assignment 3 - Exponential, Logarithmic and Trigonometric Applications

Student name _____	Date _____
Total marks 45	Suggested due date October 2, 2026 at 11:59 p.m. ET

- solve exponential and logarithmic equations with valid-domain checks
- interpret periodic parameters in sinusoidal models
- evaluate assumptions and units in applied function models

Submission instructions

Use exact values when practical and round final measured quantities to the requested precision. Define every variable and include units. Check logarithmic domains before accepting solutions.

Submit one readable PDF through the matching Moodle assignment. Include your name, page numbers, and complete reasoning. Do not submit shared links or temporary draft-file URLs.

1. A savings balance is modelled by $A(t) = 1800(1.045)^t$, where t is years. Find $A(6)$, then use logarithms to determine when the balance first exceeds \$2500. /8

2. Solve $\log_2(x - 1) + \log_2(x + 1) = 3$. State the domain and reject any invalid roots. /7

3. A medication amount follows $M(t) = 120(0.82)^t$. Find the half-life to the nearest tenth of an hour and interpret the answer. /7

4. For $y = 3\cos(2(x - \pi/4)) - 1$, state amplitude, period, phase shift, midline, maximum, and minimum. Sketch one cycle. /8

5. Daylight is modelled by $D(t) = 3.1\sin[(2\pi/365)(t - 80)] + 12.2$. Interpret every parameter and predict daylight on day 172. /8

6. Compare one strength and one limitation of exponential and sinusoidal models in real Canadian contexts. /7

Completion check

Before submitting: confirm every restriction is stated, graphs are labelled, units are included, and the total PDF is readable at 100% zoom.